Chapter 3, Section 10

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1. Over a nonperfect field, smooth and regular are not equivalent. For example, let k_0 be a field of characteristic $p > 0$, let $k = k_0(t)$, and let $X \subseteq \mathbf{A}^2_k$ be the curve defined by $y^2 = x^p - t$. Show that every local ring of X is a regular local ring, but X is not smooth over k.

Proof. To show every local ring of X is a regular local ring, it suffices to show the coordinate ring of X defined by $A = k[x, y]/(y^2 - x^p - t)$ is integrally closed. If $p \neq 2$, the result follows from (II, Ex. 6.4), so assume $p = 2$. An immediate consequence is

$$y^2 - x^2 - t = (x + y)^2 - t,$$

which implies the isomorphism

$$A = \frac{k[x, y]}{(x + y)^2 - t} \cong k(\sqrt{t})[z].$$

But $k(\sqrt{t})$ is a field, and polynomial rings over fields are integrally closed, so we conclude that A is integrally closed.

Let $\bar{k}$ be the algebraic closure of k. By (10.2), X is smooth over k if and only if $X \times_k \bar{k}$ is regular, i.e., $A \otimes_k \bar{k}$ defines a nonsingular $\bar{k}$-variety; but $t^{1/p} \in \bar{k}$, so we can factorize

$$y^2 - x^p - t = y^2 - (x + t^{1/p})^p.$$

Thus, $(-t^{1/p}, 0)$ is a singular point of $X \times_k \bar{k}$ (A.M., Ex. 11.1), so X is not smooth over k. □

2. Let $f : X \to Y$ be a proper, flat morphism of varieties over k. Suppose for some point $y \in Y$ that the fiber X_y is smooth over $k(y)$. Then show that there is an open neighborhood V of y in Y such that $f : f^{-1}(V) \to V$ is smooth.

Proof. Flat morphisms of finite type between Noetherian schemes are open (Ex. 9.1), proper morphisms are closed, X and Y are irreducible, so f must be surjective, and the generic point $\xi \in X$ is mapped to the generic point of Y. Some immediate consequence are the following: f trivially satisfies (2) in the definition of smoothness of relative dimension $n = \dim X - \dim Y$, so for every open set $V \subseteq X$, $f : f^{-1}(V) \to V$ satisfies (2). Thus, we want to find an open neighborhood V of y such that $\Omega_{X/Y}$ is free at every point in $f^{-1}(V)$ (10.0.2). For every $x \in X_y$, we have the inequalities

$$\dim X - \dim Y \leq \dim_{k(\xi)} \Omega_{X/Y} \otimes k(\xi) \leq \dim_{k(x)} \Omega_{X/Y} \otimes k(x) = \dim X_y,$$

by (II, Ex. 5.8a) and (II, 8.6A), where the last equality is by smoothness of X_y over $k(y)$ and the fact that relative differentials commute with base change. Since $n = \dim X_y$ (9.6), $\Omega_{X/Y}$ is free of rank n at every $x \in X_y$ (II, 8.9).

Now, let $U = \{x \in X \mid \Omega_{X/Y} \text{ is free at } x\}$, and $Z = X - U$. Being free is an open property for finitely generated modules, so U is an open subset of X. Then f is an open and closed mapping, so $f(U)$ and $X - f(Z)$ are open sets in Y. Also, $f^{-1}(y) = X_y$ is a subset of U, which means y is not in $f(Z)$. Therefore, $f(U) \cap (X - f(Z))$ is a nonempty open set containing y, so we can find an open neighborhood V of y such that $V \subseteq f(U) \cap (X - f(Z))$. Since $f^{-1}(V) \subseteq U$, $\Omega_{X/Y}$ is free at every point in $f^{-1}(V)$, which is what we wanted to show. □

3. A morphism $f : X \to Y$ of schemes of finite type over k is *étale* if it is smooth of relative dimension 0. It is *unramified* if for every $x \in X$, letting $y = f(x)$, we have $\mathfrak{m}_y \cdot \mathcal{O}_{x,X} = \mathfrak{m}_x$, and $k(x)$ is a separable algebraic extension of $k(y)$. Show that the following conditions are equivalent:

(i) f is étale;

(ii) f is flat, and $\Omega_{X/Y} = 0$;

(iii) f is flat and unramified.

Proof. $(i) \iff (ii)$ The forward direction is trivial. On the other hand, $\Omega_{X/Y} = 0$ trivially satisfies condition (3) in the definition of smoothness, and we want to show condition (2) holds as well. According to (9.6), condition (2) is equivalent to saying that every irreducible component of every fibre X_y of f has dimension 0, so let X' be an irreducible component of X_y with its reduced induced structure. Then X' is an integral scheme of finite type over $k(y)$, and $\Omega_{X'/k(y)}$ is a coherent sheaf, so by (II, Ex. 3.20b) and (II, 8.6A), we reduce to showing $\Omega_{X'/k(y)} \otimes k(x) = 0$ for every $x \in X'$. Then

$$\Omega_{X_y/k(y)} \otimes k(x) = \Omega_{X/Y} \otimes k(x) = 0$$

since relative differentials commute with base extension and $\Omega_{X/Y} = 0$. Hence, the result follows from the surjective map of (II, 8.12)

$$\Omega_{X_y/k(y)} \otimes k(x) \longrightarrow \Omega_{X'/k(y)} \otimes k(x) \longrightarrow 0.$$

$(i) \iff (iii)$ Relative differentials commute with base extension, so by the equivalence of (i) and (ii), we reduce to the case when $Y = \operatorname{Spec} k$. Thus, we want to show X is smooth of relative dimension 0 over k if and only if $\mathfrak{m}_x = 0$ and $k(x)$ is a separable algebraic extension of k for every $x \in X$. Consider the exact sequence of (II, 8.4A)

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{X/k} \otimes k(x) \longrightarrow \Omega_{k(x)/k} \longrightarrow 0.$$

In the forward direction, suppose $\Omega_{X/k} = 0$. Then $\Omega_{k(x)/k} = 0$ by the exact sequence, so $k(x)$ is a separable algebraic extension of k (II, 8.6A) Also, X is smooth of relative dimension 0 over k if and only if $\bar{X} = X \times \bar{k}$ is equidimensional of dimension zero and regular, where $\bar{k}$ an algebraic closure of k (10.2). Regularity and dimension of $X \times \bar{k}$ implies $\mathfrak{m}_x \otimes_k \bar{k} = 0$, hence $\mathfrak{m}_x = 0$. On the otherhand, if $\mathfrak{m}_x = 0$ and $k(x)$ is a separable algebraic extension of k. Then $\Omega_{k(x)/k} = 0$ (II, 8.6A), and the term on the left hand of the exact sequence is zero. Hence, $\Omega_{X/k} = 0$. □

5. If x is a point of a scheme X, we define an *étale neighborhood* of x to be an étale morphism $f : U \to X$, together with a point $x' \in U$, such that $f(x') = x$. As an example of the use of étale neighborhoods, prove the following: if $\mathcal{F}$ is a coherent sheaf on X, and if every point of X has an étale neighborhood $f : U \to X$ for which $f^*\mathcal{F}$ is a free $\mathcal{O}_U$-module, then $\mathcal{F}$ is locally free on X.

Proof. Consider the following lemma:

Lemma 1. *If B is a faithfully flat A-algebra, then for any A-module M, we have M is A-flat if and only if $M \otimes_A B$ is B-flat.*

Let $\mathfrak{a}$ be an ideal of A. Then

$$\operatorname{Tor}_1^A(M, A/\mathfrak{a}) \otimes_A B \cong \operatorname{Tor}_1^B(M \otimes_A B, B/\mathfrak{a}B) = 0,$$

so by faithfull flatness, we conclude that $\operatorname{Tor}_1^A(M, A/\mathfrak{a}) = 0$ (A.M., Ex. 3.16).

Now, if $f^*\mathcal{F}$ is a free $\mathcal{O}_U$-module, then $(f^*\mathcal{F})_{x'}$ is a free $\mathcal{O}_{x',U}$-module with $(f^*\mathcal{F})_{x'} \cong \mathcal{F}_x \otimes_{\mathcal{O}_{x,X}} \mathcal{O}_{x',U}$, and $\mathcal{O}_{x',U}$ is a faithfully flat $\mathcal{O}_{x,X}$-algebra, so the result follows from Lemma 1 and the fact that for finitely generated modules over Noetherian local rings, being free is equivalent to being flat. □

6. Let Y be the plane nodal cubic curve $y^2 = x^2(x+1)$. Show that Y has a finite étale covering X of degree 2, where X is a union of two irreducible components, each one isomorphic to the normalization of Y.

Proof. For simplicity, assume k is algebraically closed and has characteristic $\neq 2$. Consider two copies of the twisted cubic curve X_1 and X_2 in $\mathbf{A}^3$ with affine coordinates x, y, z given by the parametric equations

$$X_1 : \begin{cases} x = t^2 - 1 \\ y = t^3 - t \\ z = t \end{cases}, \quad X_2 : \begin{cases} x = t^2 - 1 \\ y = t^3 - t \\ z = -t \end{cases}.$$

Each are isomorphic to the normalization of Y, and they intersect at the following three points

$$P_1 = (0,0,1), \quad P_2 = (0,0,-1), \quad Q = (-1,0,0).$$

Let X be the closed subscheme of X with two irreducible components X_1 and X_2. There is an obvious map $X \to Y$ given by the projection map onto the xy-plane, and each irreducible component of X maps onto Y. Finally, let $\pi : \tilde{X} \to X$ be obtained from X by blowing up the point Q. Then $\tilde{X} \to Y$ is our desired étale covering of degree 2. □